BhuMi

P4 / SMART E-MUTATION & REGISTRY

Land Ownership Reconciliation Engine

Every land transaction should leave a traceable, consistent ownership record.

DOCUMENTED INTEREST

1/2

TRANSFERRED INTEREST

1/4

REMAINING INTEREST

1/4

Team BhuMi · Daffodil International University, Bangladesh

Digital records can still disagree.

A deed and its mutation can add up — while assigning land interests to the wrong people.

01

Registered deed

Records a transfer between specific parties.

02

Ownership history

Shows what the transferor held and what co-owners retain.

03

Mutation record

May omit or misallocate the resulting shares.

The hidden risk: an uninvolved co-owner loses interest even when the total stays correct.

Context: World Bank, Bangladesh Land Acquisition Diagnostic Review (2022). Prototype examples are fictional.

Reconcile ownership, not just documents.

BhuMi independently reconstructs the expected ownership state, then checks each party against mutation.

Exact Fractional shares, not rounded decisions.

Explainable Named parties, exact differences and source IDs.

Reviewable An officer records the next step and the reason.

WHAT MAKES BHUMI DIFFERENT

Correct total area

≠

Correct ownership allocation

A MATCH means the supplied records reconcile. It does not establish legal title.

From documented history to a reviewable result.

01

Ownership history

Baseline + sourced events

›

02

Expected shares

Replay exact allocations

›

03

Mutation comparison

Check every owner

›

04

Evidence & review

Explain, review, retain

Evidence comes before certainty.

Missing sources or unclear event order prevent a false MATCH.

Keep three records separate

Baseline · Events · Mutation

A mutation never overwrites the expected history.

100 decimals. A hidden 10-decimal shift.

Fictional Rahim holds 60; Karim holds 40. Rahim sells 25 to fictional buyer Nabila.

PARTY	EXPECTED	MUTATION	DIFFERENCE
Rahim	35	35	0
Karim	40	30	−10
Nabila	25	35	+10
TOTAL	**100**	**100**	**0**

Both totals balance. Karim loses 10 without participating in the sale.

Run DEMO-P4-011 · MUTATION_WRONG_SHARE · All figures are decimals equivalent of undivided interests.

Built around evidence an officer can inspect.

Exact share accounting

BigInt fractions keep thirds, sixths and transfers exact.

Nine focused rules

Over-transfer, missing mutation, wrong shares and broken chains.

Source-linked evidence

Expected versus actual shares with event and document references.

Review & audit trail

Request evidence, resolve or escalate with a recorded reason.

Corrected inputs create a new result. Earlier findings and review actions remain visible.

A transparent engine. A tested prototype.

INTERFACE

Next.js · React · Tailwind

DOMAIN

TypeScript · Zod · exact BigInt fractions

STORAGE

PostgreSQL via PGlite · Drizzle ORM

27	unit tests passed
3	browser tests passed
11	synthetic scenarios checked

Also verified: production build, type checks, lint and persistence across a server restart.

Local measured results, not national accuracy claims. PGlite is a single-instance embedded PostgreSQL prototype.

Support land offices with clearer exceptions.

Potential value: help reviewers focus on the exact party, share and source that need attention.

NOW

Synthetic prototype

Demonstrate reproducible discrepancies and a traceable review workflow.

NEXT

Authorized pilot

Validate rules with land and legal experts using permitted data extracts.

LATER

Institutional integration

Add hosted storage, officer identities, bilingual UX and approved interfaces.

Complement existing e-mutation infrastructure. Do not replace official or judicial decisions.

Digital records should tell the same ownership story.

BhuMi shows where they disagree — and gives a human reviewer the evidence.

STUDENT 1 / TEAM LEAD

[Name · DIU ID]

Engine, data model and tests

STUDENT 2 / DEVELOPER

[Name · DIU ID]

Interface, API and demonstration

Faculty mentor: [Name · Department] Industry mentor: [Name · Organisation]

github.com/void-pronway/BhuMi-Land-Reconciliation

AI assistance: OpenAI ChatGPT (GPT-6) and Codex. Student review and ownership remain essential.